Number Theory — problem 6.4

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(a) $3 \mid a$, $3 \mid b \Leftrightarrow 3 \mid a^2 + b^2$. (b) $7 \mid a$, $7 \mid b \Leftrightarrow 7 \mid a^2 + b^2$. (c) $21 \mid a^2 + b^2 \Rightarrow 441 \mid a^2 + b^2$.